

Appendix: Using Hierarchical Models to Estimate Heterogeneous Effects

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1 Replication Analysis Details

To produce Figure 1 in the manuscript, I downloaded all replication files from the Harvard Dataverse for papers in the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics*. The analysis covers from 2010 to 2024 because of lags in replication data upload and approval for 2025 and 2026. This also means that the analysis covers the year the replication data was uploaded, not the year of publication. This is important because publication often lags acceptance by a year.

I then identified code files in the replication materials, downloading scripts in R, Stata, Python and other coding languages. Within those files, I used regular expressions to identify models with an interaction, uses of associated tools such as marginal effects or STATA's margins command, and machine-learning tools for heterogeneous effects. I also parsed comments. The two lists below summarize regular expressions for interactions and machine-learning.

Interaction Patterns

- `(lm|glm|felm|feols|plm|ivreg)\s*\([^)]*\s*~\^[^)]*\s*[:]`
- `feols?\s*\([^)]*\s*|`
- `[ci]\.\w+#`
- `\w+##\w+`
- `dydx\s*\s*(`
- `margins\s*,.*at\s*\s*(`
- `margineffects|slopes\s*\s*(|comparisons\s*\s*(`
- `heterogeneous|eity)\s*(effect|treatment)?`
- `conditional\s*(average)?\s*(treatment)?\s*effect`
- `subgroup\s*analysis|stratified\s*analysis`
- `moderation|ing|or)\s*(effect|analysis|variable)?`
- `interaction\s*(effect|term|model)`
- `interplot|interact_plot|marginsplot|coefplot.*#`

ML Heterogeneity Patterns

- `causal_forest\s*\ (`
- `library\s*\ (\s*grf\s*\) | require\s*\ (\s*grf\s*\)`
- `average_treatment_effect|predict.*causal_forest`
- `bartc\s*\ (|library\s*\ (\s*bartCause\s*\)`
- `bcf\s*\ (|library\s*\ (\s*bcf\s*\)`
- `\bCATE\b|conditional\s+average\s+treatment\s+effect`
- `\bITE\b|individual(ized)?\s+treatment\s+effect`
- `(causal|treatment)\s*(forest|tree|learning)`
- `DoubleML|double.*machine.*learning|dml_`
- `[STXR] [-_] ?learner|metalearner`

2 Simulation Data Generating Process

This section summarizes the data generating process for the simulations in the main text. Both simulations fix the sample size at $n = 2000$ and vary the number of groups, with different numbers of observations in each group.

2.1 Equal Group Size Simulation

For the first simulation, I generate k binary grouping variables, where $k \in \{3, 4, 5, 6, 7\}$:

$$x_{ij} \sim \text{Bernoulli}(0.5) \quad \text{for } j = 1, \dots, k$$

The simulation defines the groups by unique combinations of these variables, producing 2^k groups. With three modifiers, there are 8 groups, while 7 modifiers produces 128 groups.

The treatment indicator is randomly assigned, also with a bernoulli draw:

$$T_i \sim \text{Bernoulli}(0.5)$$

Three control variables predict the baseline outcome but do not modify the treatment effect:

$$z_{i1} \sim N(0, 1)$$

$$z_{i2} \sim \text{Bernoulli}(0.5)$$

$$z_{i3} \sim N(0, 1)$$

True group-specific treatment effects are drawn from a normal distribution with mean μ_τ and variance σ_τ , both of which are fixed:

$$\tau_g \sim N(\mu_\tau, \sigma_\tau) \quad \text{where } \mu_\tau = 0.2, \sigma_\tau = 0.3$$

The outcome is then generated as:

$$y_i = \beta_1 z_{i1} + \beta_2 z_{i2} + \beta_3 z_{i3} + \tau_{g[i]} T_i + \epsilon_i$$

$$\epsilon_i \sim N(0, \sigma_y) \quad \text{where } \sigma_y = 1$$

I also fix the control coefficients $\beta = (0.4, -0.3, 0.25)$.

The third simulation replaces the group treatment effects draw with a student-t distribution, using either 3 or 5 degrees of freedom.

$$\tau_g \sim \text{Student T}(df, \mu_\tau, \sigma_\tau) \quad \text{where } df = (3, 5), \mu_\tau = 0.2, \sigma_\tau = 0.3$$

2.2 Unequal Group Size Simulation

The second simulation uses the same data generating process but fixes $k = 6$ (64 groups) and varies the probability parameter for the grouping variables. Rather than fixing all probabilities at 0.5, I vary them symmetrically across groups to create imbalanced group sizes. I consider four scenarios:

1. Balanced (all probabilities at 0.5)
2. Mild imbalance (0.35, 0.4, 0.5, 0.5, 0.6, 0.65)
3. moderate imbalance (0.25, 0.35, 0.5, 0.5, 0.65, 0.7)
4. Strong imbalance (0.2, 0.3, 0.5, 0.5, 0.675, 0.75)

At maximum imbalance, some groups may have as few as one observation while others are much larger. I set the simulation so that all groups have at least one observation.

3 Tomz and Weeks Reanalysis

This section summarizes the hierarchical models of Tomz and Weeks (2021) and details the underlying R code. Both models use the nonlinear lambda specification in brms, which separates the outcome into two linked equations: a heterogeneity equation modeling the treatment effect (λ) and a control equation.

3.1 R Code: Tomz and Weeks Reanalysis

First, the following model formula expresses a heterogeneous treatments model where the impact of alliances varies with the experimental conditions (regime, stakes, costs, and region). The `treat_group` indicator marks unique combinations of these experimental factors for the group-level random intercept. The control equation includes respondent demographics and foreign policy dispositions.

```
bf(
  force ~ lambda * alliance + controls,

  lambda ~ regime + stakes + costs + region_txt + (1 | treat_group),

  controls ~ white + male + hawk + intl + dem + rep + age + ed4,

  nl = TRUE
)
```

The second formula expresses a respondent heterogeneity model where the impact of alliances varies with demographic characteristics (party, news interest, race, and gender). The `het_group` indicator marks unique combinations of these demographic modifiers. The control equation includes experimental conditions and respondent age and education.

```
bf(
  force ~ lambda * alliance + controls,

  lambda ~ (dem + rep + high_newsint + white + male) + (1 | het_group),

  controls ~ regime + stakes + costs + africa + europe + asia + age + ed4,

  nl = TRUE
)
```

3.2 Priors: Tomz and Weeks Reanalysis

Table 1 summarizes the prior distributions for both models. All priors are weakly informative relative to the scale of the data. The lambda coefficients receive a $N(0, 1)$ prior, while

control coefficients receive a tighter $N(0, 0.5)$ prior.

Parameter	Prior
λ coefficients	$N(0, 1)$
Control coefficients	$N(0, 1)$
σ (residual SD)	half- $N(0, 1)$
σ_θ (group SD)	half- $N(0, 1)$

Table 1. *Summary of priors in hierarchical models.*

4 Bush and Prather Reanalysis

In the following, I again demonstrate how the model works by reanalyzing a study by Bush and Prather (2020) (BP hereafter). This study examines how foreign meddling in elections impacts support for economic engagement with the meddler. One of their experiments examines how Russian or German engagement in the 2016 US election impacts mass support for trade and investment with those countries. I use a heterogeneous effects model to check their results and further explore their findings.

BP employ a 2x2x2 factorial experiment. This design randomizes whether a foreign country is interfering in the 2016 election, the country and their attitude towards Trump and whether the potential economic ties entail greater trade or investment in the United States. In one meddling treatment, Russia expresses support for Trump and in the other, Germany expresses opposition to Trump.

BP hypothesize that individuals will prefer economic engagement with states that support their candidate. They thus examine how the impact of side-taking varies with the direction of the endorsement, individual political preferences, and the economic issue. To do this, they use four tables and figures, and rely on eight t-tests of differences in means between groups of roughly 25 respondents.

I use the same nonlinear lambda specification as in the Tomz and Weeks reanalysis. The heterogeneous effects model estimates the impact of side-taking on economic engagement, with the treatment effect (λ) modeled as a function of potential modifiers. The heterogeneity equation includes a dummy indicator of intention to vote for Clinton, an indicator of whether the experiment deals with trade or investment, female gender, and high political engagement. I measure political engagement as a sum of political knowledge and interest, dichotomized at the median. Groups are defined by unique combinations of these four binary modifiers, producing 16 groups.

The outcome variable is a scale from one to four that measures support for greater economic engagement with the potential partner. To approximate BP's analyses, I use a normal outcome likelihood. The control equation includes college education and employment status.

4.1 R Code: Bush and Prather Reanalysis

The following model formula expresses the heterogeneous effects model:

```
bf(
  ipe_support ~ lambda * treat_germrus + controls,

  lambda ~ (w2_vote_hill + woman + high_pol_engage + invest_cond) +
    (1 | het_group),

  controls ~ college_educ + employ_dum + woman + high_pol_engage

  nl = TRUE
```

)

The priors follow the same structure as the Tomz and Weeks models, with $N(0, 1)$ on coefficients in the heterogeneity equation and $N(0, 0.5)$ on control coefficients.

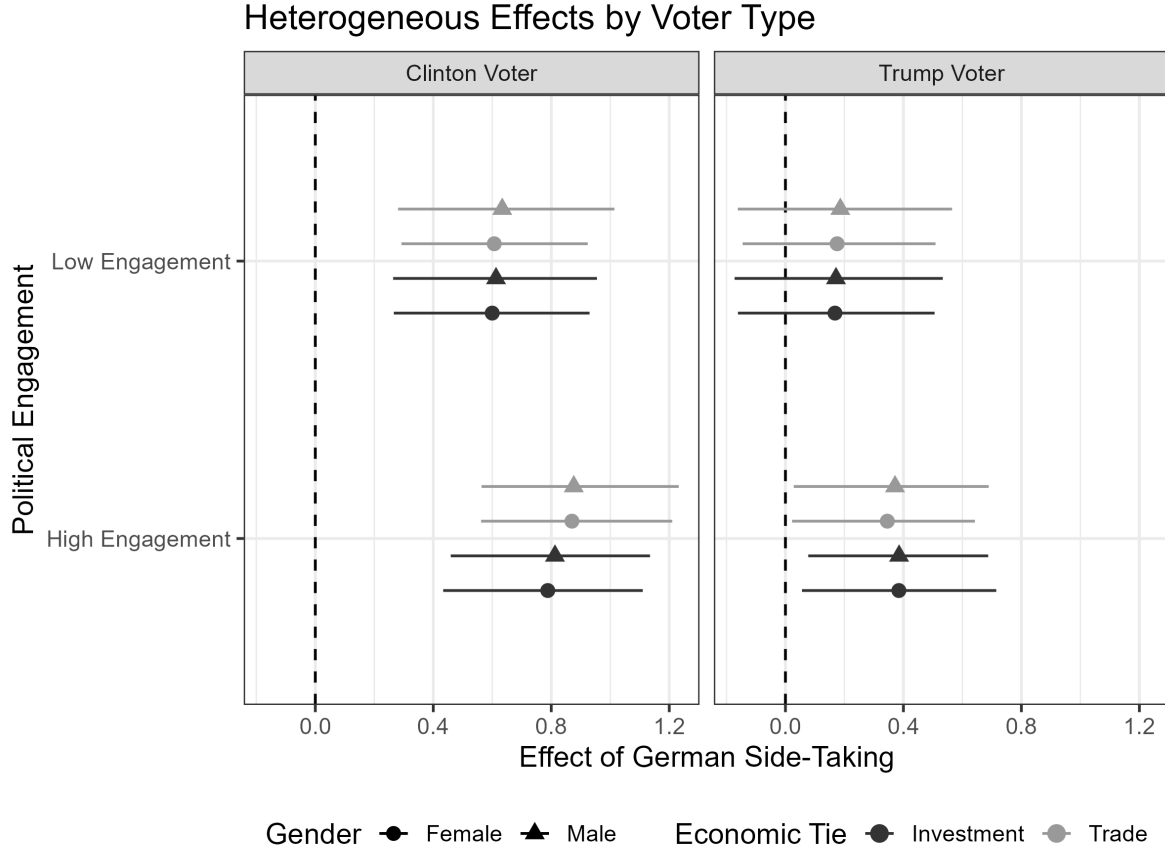


Figure 1. *Heterogeneous effects of German side-taking in US elections on support for economic engagement with Germany. Each estimate reflects a treated group with a unique combination of other treatments and demographic characteristics.*

These estimates corroborate BP's findings. Clinton voters are more likely to support economic engagement with Germany when Germany has opposed Donald Trump. This has the largest impact on attitudes towards trade among highly engaged respondents. Low engagement Clinton voters are less responsive.

Trump voters are less responsive to German opposition to Trump. The effects are mostly null, and only for engaged voters in the investment condition does the endorsement have a clearly non-zero effect. This positive effect may reflect residual opposition to Trump in some sections of the Republican party.

References

- Bush, Sarah Sunn and Lauren Prather. 2020. "Foreign Meddling and Mass Attitudes Toward International Economic Engagement." *International Organization* 74(2):584–609.
- Tomz, Michael and Jessica L.P. Weeks. 2021. "Military Alliances and Public Support for War." *International Studies Quarterly* 65(3):811–824.